Project Planning & Management

Smart Healthcare Platform (Shefaa - شفاء)

DEPI Graduation Project – October 2026

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Contents

1. Project Proposal
2. Project Plan
3. Task Assignment & Roles
4. Risk Assessment & Mitigation Plan
5. Key Performance Indicators (KPIs)

1. Project Proposal

1.1 Project Overview

The Smart Healthcare Platform is a web-based medical system that connects patients, doctors, and laboratories in one secure place. Each patient owns a digital medical record and decides which doctors can view it. Doctors write diagnoses and electronic prescriptions, laboratories upload test results directly to the patient's file, and an administrator manages accounts without access to any medical data.

The platform focuses on privacy and permission control, and adds an interactive map, an analytics dashboard for doctors, and an AI assistant that suggests the suitable medical specialty based on the patient's symptoms.

1.2 Problem Statement

- Patients' medical records are scattered across clinics, labs, and paper files, which makes follow-up and sharing difficult.
- Patients usually have no control over who can see their medical history.
- Finding a suitable doctor or nearby laboratory takes time, and patients often do not know which specialty fits their symptoms.
- Medical data is highly sensitive and needs strong protection against leaks and unauthorized access.

1.3 Objectives

- Give each patient a single digital medical record under their full control.
- Let doctors access a patient's history only after the patient's permission.
- Allow laboratories and radiology centers to upload results directly to the patient's file.
- Protect all sensitive data using encryption in transit and at rest.
- Help patients find the right specialty and available doctors through an AI-assisted suggestion tool and an interactive map.
- Provide doctors with clear analytics about their patients' conditions.

1.4 Scope

In scope:

Five user roles: Patient, Doctor, Laboratory / Radiology Center, Receptionist, and System Administrator.

Role-based access control with patient-managed access permissions.

Appointment booking, electronic prescriptions, diagnoses, and lab result uploads.

Receptionist pages:

Reception dashboard showing today’s queue and each patient’s status.

Book, edit, or cancel appointments on behalf of patients.

Patient check-in.

View administrative patient data only (name, phone number), with no medical data.

Request access permission from an in-person patient using OTP or QR.

Record invoices and payment status (record keeping only, no payment gateway).

Interactive search map for doctors and laboratories.

Doctor analytics dashboard with charts.

AI chatbot / symptom form that suggests a specialty and available doctors.

Doctor license verification by the administrator.

Out of scope:

The AI assistant does not diagnose diseases or recommend treatments, to avoid medical and legal liability.

Online payment gateways and video consultations (may be considered as future work).

1.5 Target Users and Access Summary

Role	Main Permissions
Patient	Full control of own medical record, books appointments, grants or revokes doctor access, uses the AI assistant and map.
Doctor	Reads patient history only when permitted, writes diagnoses and electronic prescriptions, views analytics dashboard.
Receptionist	Manages today's queue, appointments, and patient check-in; sees name and phone only; requests access permission from the patient via OTP/QR; records invoices. Cannot view any medical data.
Laboratory / Radiology	Uploads results and reports to the patient's file; cannot edit prescriptions or view unrelated history.
Administrator	Manages accounts, verifies doctors' licenses, monitors system performance; cannot view patients' medical data.

1.6 Key Features

- **Role-based system:** Patient, doctor, laboratory, and admin dashboards with protected routes.
- **Security:** HTTPS for data in transit and encryption of medical data at rest.
- **Interactive map:** Search for doctors and laboratories using a map (Google Maps API or Leaflet.js).
- **Doctor analytics:** Most common diseases, patient adherence to medication, and progress of a patient's condition over time (Chart.js or Recharts).
- **Reception desk**: Daily patient queue, appointment management, patient check-in, and access permission requests through OTP or QR. The receptionist sees administrative data only.
- **AI assistant:** Symptom-based specialty suggestion and list of available doctors.

1.7 Proposed Architecture

The system is planned as a microservices (or hybrid cloud) architecture to separate permissions and improve security:

- User interface for Patient, Doctor, Laboratory, and Receptionist.
- Security gateway.
- Identity and Permissions Service (users database).

- Reception & Scheduling Service (appointments, daily queue, check-in, and invoice records). It stores administrative data only (name, phone) and has no access to medical data.
- Medical Data Service (encrypted database and files).

The final technology stack will be confirmed during the System Analysis & Design phase.

1.8 Team

#	Team Member	Role / Responsibility
1	Bimen Hanna	Patient & Reception Module
2	Bavly Hany	Doctor Module
3	Aml Mahrous	Laboratory, Admin & Security
4	Mona Mohamed	AI, Map & Documentation

1.9 Project Timeline

The plan follows the official DEPI deadlines and divides the work into four phases:

Phase	Period	Main Activities
1. Planning & Requirements	11 – 16 Oct 2026	Project proposal, project plan, task assignment, risk assessment, KPIs, stakeholder analysis (including the receptionist), user stories and use cases, functional and non-functional requirements. Upload all documents to GitHub.
2. System Analysis & Design	17 Oct – 6 Nov 2026	Use case and sequence diagrams, database design (ERD), system architecture, roles and permissions matrix, AP Idesign, and UI wireframes for all five roles (including the reception dashboard and check-in flow).
3a. Implementation – Foundation	7 – 13 Nov 2026	Project setup, database, sign in / sign up, role-based access control (five roles), and protected routes.
3b. Implementation – Core Features	14 – 20 Nov 2026	Patient pages (medical record, appointment booking, access permissions), reception pages (daily queue, appointment management, patient check-in, OTP/QR access request), doctor pages (patients, diagnoses, prescriptions), and laboratory result upload.
3c. Implementation – Advanced Features	21 – 27 Nov 2026	Interactive map, doctor analytics dashboard, AI symptom assistant, admin panel (accounts and license verification), data encryption, and invoice records for the receptionist (optional, if time allows).
3d. Integration & Bug Fixing	28 – 30 Nov 2026	Connect all parts together, fix bugs, and prepare the final version of the source code.
4. Testing, Reports & Presentation	1 – 4 Dec 2026	Final testing, final report, documentation, and presentation rehearsal.

Deadlines: Planning and Requirements on 16 Oct, Design on 6 Nov, Implementation on 30 Nov, and Final Presentation on 4 Dec 2026.

1.10 Expected Deliverables

- Planning documents: Proposal, Project Plan, Task Assignment, Risk Assessment, KPIs.
- Requirements documents: Stakeholder Analysis (including the receptionist), User Stories & Use Cases, Functional and Non-functional Requirements.
- System analysis and design documents (including the permissions matrix for all five roles).
- Working web application with five role-based dashboards and source code on GitHub.
- Testing results, final report, and presentation.

2. Project Plan

2.1 Overview

This plan describes the timeline, milestones, deliverables, and resources of the Smart Healthcare Platform project. The project runs from 11 October 2026 to 4 December 2026 and follows the official DEPI deadlines.

2.2 Milestones and Deliverables

Milestone	Deadline	Deliverables
Planning & Requirements	16 Oct 2026	Proposal, Project Plan, Task Assignment, Risk Assessment, KPIs, Stakeholder Analysis, User Stories & Use Cases, Functional and Non-functional Requirements
System Analysis & Design	6 Nov 2026	Use case and sequence diagrams, database design (ERD), architecture, permissions matrix, API design, UI wireframes
Implementation	30 Nov 2026	Working web application and source code on GitHub
Final Presentation, Testing & Reports	4 Dec 2026	Test results, final report, and presentation

2.3 Gantt Chart

Phase	11–16 Oct	17–23 Oct	24–30 Oct	31 Oct–6 Nov	7–13 Nov	14–20 Nov	21–27 Nov	28 Nov–4 Dec
Planning & Requirements	■							
System Analysis & Design		■	■	■				
Implementation – Foundation					■			
Implementation – Core Features						■		
Implementation – Advanced Features							■	
Integration & Bug Fixing								■
Testing, Reports & Presentation								■

Each shaded cell means the phase is active during that week.

2.4 Implementation Breakdown

Stage	Period	Main Work
Foundation	7 – 13 Nov	Project setup, database, sign in / sign up, role-based access control (five roles), protected routes, base layout and empty dashboards for all five roles

Stage	Period	Main Work
Core Features	14 – 20 Nov	Patient pages, reception pages (daily queue, appointment management, patient check-in, OTP/QR access request), doctor pages (patients, diagnoses, prescriptions), laboratory result upload
Advanced Features	21 – 27 Nov	Interactive map, doctor analytics dashboard, AI symptom assistant, admin panel (accounts and license verification), data encryption, invoice records for the receptionist (optional, if time allows)
Integration & Bug Fixing	28 – 30 Nov	Connect all parts, fix bugs, final version of the code on GitHub

2.5 Resource Allocation

Team (4 members): Bimen Hanna, Bavly Hany, Aml Mahrous, Mona Mohamed. Responsibilities of each member are defined in the Task Assignment & Roles document.

Tools (to be finalized in the design phase):

- GitHub for code and documents, with a separate branch for each member.
- React for the user interface; Chart.js or Recharts for charts; Leaflet.js or Google Maps API for the map.
- Figma (or similar) for UI wireframes.
- Word / PDF for documentation.

3. Task Assignment & Roles

3.1 Team Members

#	Team Member	Main Role
1	Bimen Hanna	Patient & Reception Module
2	Bavly Hany	Doctor Module
3	Aml Mahrous	Laboratory, Admin & Security
4	Mona Mohamed	AI, Map & Documentation

3.2 Work Packages

The project is divided into four work packages. Each package is owned by one team member, and members help each other when needed.

#	Work Package	Scope	Assigned To
1	Patient Module	Dashboard, Medical Record, Prescriptions & Medications, Results, Book Appointment, My Appointments, Permission Management, Settings, Reception Dashboard, Appointment Booking & Editing, Check-in, Patient Administrative Data, Access Permission Request (OTP/QR),	Bimen
2	Doctor Module	Dashboard, Schedule & Working Hours, My Patients List, Access Request, Patient File, Examination Page, Case Progress, Clinic Team Management, Professional Profile	Bavly
3	Laboratory, Admin & Security	Lab Dashboard, Order Details, Result Upload, Results History, Lab Profile, Account Management, License Verification, Monitoring & Reports	Aml
4	AI, Map & Documentation	Home, Map, Public Doctor & Lab Profiles, AI Symptom Triage, About / Privacy / Terms, Documentation, Report, Presentation	Mona

3.3 Shared Responsibilities

Responsibility	Description	Owner
GitHub repository management	Merge pull requests, keep the main branch clean	Mona
Code review	Each pull request is reviewed by another member	All members
Testing	Each member tests own module; final testing together	All members
Weekly progress meeting	Review progress, blockers, and next tasks	All members

3.4 Working Rules

- Every member works on a separate Git branch and never pushes directly to the main branch.
- Changes are merged through pull requests after review.
- Members pull the latest changes before starting work to avoid conflicts.
- Blockers are reported to the team as soon as they appear.

4. Risk Assessment & Mitigation Plan

4.1 Introduction

This document lists the main risks that may affect the project and the actions planned to reduce their effect. Likelihood and impact are rated Low, Medium, or High.

4.2 Risk Register

ID	Risk	Likelihood	Impact	Mitigation
R1	Leak of, or unauthorized access to, medical data	Medium	High	Encryption in transit and at rest, role-based access control, hashed passwords, security testing
R2	Delay because of the short implementation period	High	High	Weekly meetings, parallel work, prioritize high-priority requirements, keep 28–30 Nov as buffer
R3	Code conflicts between team members on GitHub	Medium	Medium	Separate branches, pull requests, pull latest changes often
R4	A team member is unavailable or overloaded	Medium	High	Clear task ownership, backup person per module, shared documentation
R5	Scope is too large (map, AI, dashboard, separate services)	High	Medium	Build core features first, add advanced features as time allows, fixed scope after design phase
R6	Problems with third-party services (map API limits, AI)	Medium	Medium	Use free options such as Leaflet with OpenStreetMap; simple rule-based specialty suggestion as a fallback
R7	AI suggestions understood as a medical diagnosis	Low	High	AI only suggests a specialty, never a diagnosis; show a clear disclaimer
R8	No real medical data available for testing	High	Low	Use fake sample data only; never use real patient data
R9	Data or code loss	Low	High	Version control on GitHub, regular database backups
R10	Integration problems between system parts	Medium	High	Define APIs early in the design phase, integrate continuously instead of at the end
R11	Lack of experience with new technologies (encryption, separate services)	Medium	Medium	Allocate learning time, simplify the architecture if needed, ask the instructor

5. Key Performance Indicators (KPIs)

5.1 Introduction

The following indicators are used to measure the success of the project. Targets are proposed and can be adjusted by the team.

5.2 KPI Table

#	Category	KPI	Target
1	Performance	Page load time	Less than 3 seconds under normal use
2	Performance	Response time of common operations (login, booking, record view)	Less than 2 seconds
3	Reliability	System uptime during testing and the final demo	99% or higher
4	Security	Unauthorized access attempts blocked in role tests	100%
5	Security	Sensitive medical data stored encrypted	100% of sensitive fields
6	Functionality	High-priority functional requirements implemented	90% or higher by 30 Nov 2026
7	Quality	Test cases passed in final testing	90% or higher, with no open critical bugs at delivery
8	Usability / Adoption	Test users who complete the main flows (book appointment, upload result, grant access) without help	At least 10 test users, 80% completion rate
9	Schedule	DEPI deadlines met	All 4 deadlines (16 Oct, 6 Nov, 30 Nov, 4 Dec 2026)